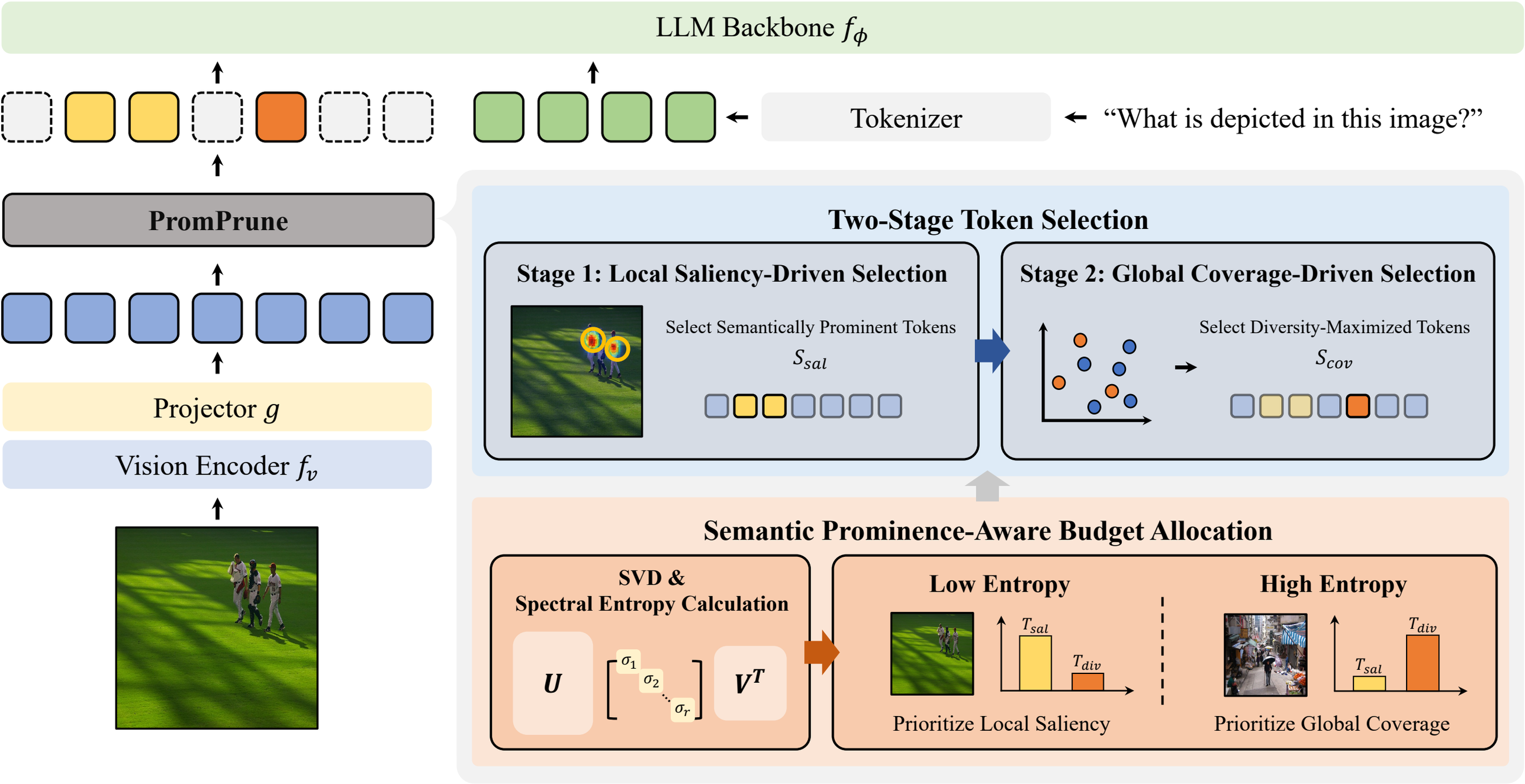
LLM Backbone f_ϕ
Tokenizer
"What is depicted in this image?"
PromPrune
Projector g
Vision Encoder f_v
Two-Stage Token Selection
Stage 1: Local Saliency-Driven Selection
Select Semantically Prominent Tokens
S_{sal}
Stage 2: Global Coverage-Driven Selection
Select Diversity-Maximized Tokens
S_{cov}
Semantic Prominence-Aware Budget Allocation
SVD &
Spectral Entropy Calculation
U
σ_1
σ_2
σ_r
V^T
Low Entropy
T_{sal}
T_{div}
Prioritize Local Saliency
High Entropy
T_{div}
T_{sal}
Prioritize Global Coverage